

Benchmarking AI Agents for Kaggle Competitions

A Fair Three-Way Comparison with Paired Significance Testing

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1 Motivation

LLM-driven agents can now solve Kaggle competitions end-to-end, but the methodology for comparing them is weak — *one run, one score, higher wins*.

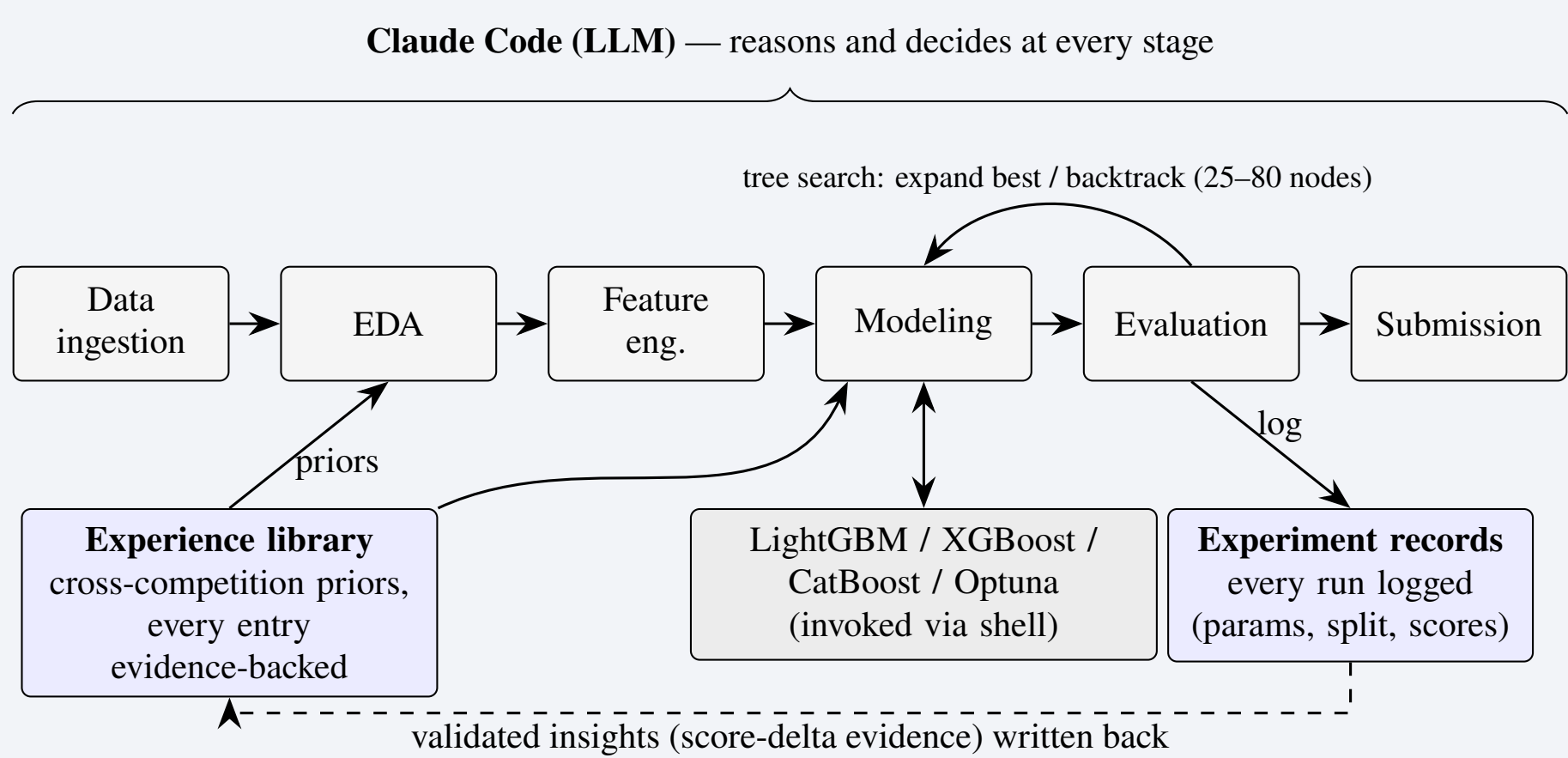
- **Q1 — Positioning.** Where does our own agent stand? Self-assessment without external, methodologically different reference points is circular.
- **Q2 — Noise.** Competition metrics carry sampling error and agent runs are stochastic. In how many competitions can agents *actually* be told apart?

2 Three distinct agents

	my-agent	AIDE	NVIDIA
Knowledge	experience library (accumulates)	none (start)	(cold public kernels)
Search	6-stage tree search, 25–80 nodes	+ tree search, 5–10 steps	tree no search, 20 produce
Validation	split by task type; diagnostics quarantined	no split legality	inherits kernel's
External data	proactively joined	never	inherited

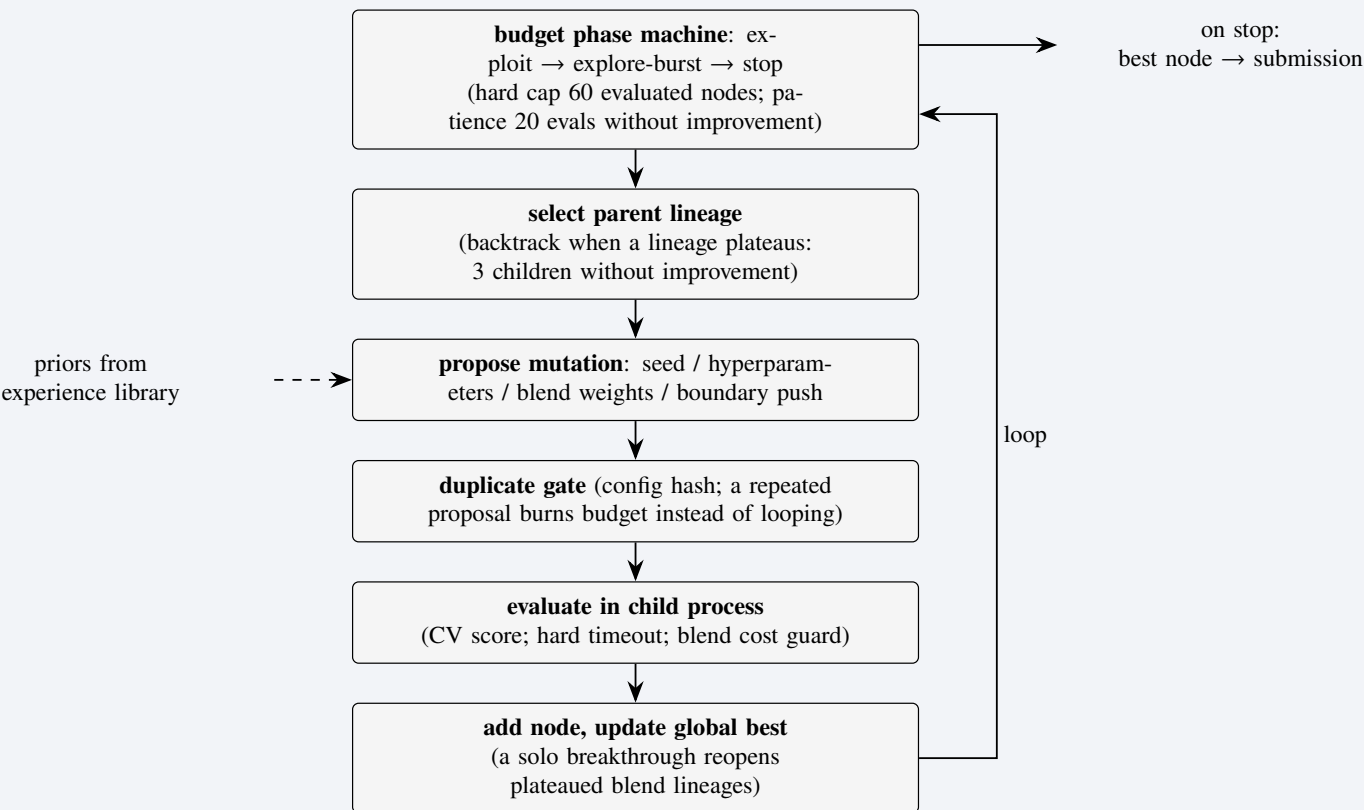
my-agent is our hybrid design (Claude Code Skill): the LLM reasons and decides; LightGBM/XGBoost/CatBoost/Optuna execute. Both reference methods are **frozen** — improving a yardstick mid-study would corrupt the comparison.

3 my-agent architecture



The experience library only stores insights carrying a *score-delta evidence field*; validated insights flow back for future competitions.

4 Tree-search loop (harness v3)



5 A fair benchmark: 18 competitions

Evaluation set. 13 Kaggle Playground competitions + 5 non-Playground (afsis 2014, conway 2014, cat-in-the-dat 2019, tps-aug/jan-2022). Inclusion criterion: both public and private leaderboard scores available. Every submission is scored on the *real* leaderboard via late submission.

Mechanically enforced fairness (three agents, one machine):

- **Data isolation** — manifest-validated isolated roots
- **Resource isolation** — mutex: one lane at a time
- **Instruction isolation** — prompts carry no recipes, no other agent's findings; environment defects are fixed, never hinted

6 The paired significance test

Kaggle scores one submission on two disjoint test subsets — two independent estimates of the same model. Their gap is that competition's *sampling noise*.

Naive thresholds fail. An agent's own public-private drift is *shared*: on s5e10 the three drifts are +0.00025/ + 0.00026/ + 0.00024 — common shift 0.00025, mutual scatter only **0.00002**. Using it as a threshold inflates noise $>10\times$.

The paired construction. For the between-agent gap d , a verdict requires:

$$\text{sign}(d_{\text{priv}}) = \text{sign}(d_{\text{pub}}) \quad \text{order replicates}$$
$$|d_{\text{priv}}| > |d_{\text{priv}} - d_{\text{pub}}| \quad \text{gap exceeds its own movement}$$

Common drift cancels in the subtraction. No extra experiments needed.

7 Results: 54 duels, 40 decidable

	Wins	Losses	Win rate
my-agent	16	11	59%
AIDE	11	13	46%
NVIDIA	13	16	45%
Ties	14 duels		(26%)

Median percentile rank vs. all human teams: my-agent **72.3**, AIDE 70.8, NVIDIA 66.7. **A quarter of all duels are statistically undecidable** — a naive tally would report that fraction as conclusions.

8 Local CV vs. the leaderboard

CV-based and leaderboard-based conclusions **disagree in 10 of 13** comparable competitions. Extreme case (s3e19, SMAPE):

Basis	my-agent	NVIDIA	Verdict
Local CV	10.02	13.72	my-agent wins big
Private LB	50.46	48.27	NVIDIA wins

The direction fully reverses: agent benchmarks must submit to the real leaderboard.

9 Ranking = method $\times$ evaluation set

Same agents, same test — swap the competition set and the ranking **fully reverses**:

Win rate on	my-agent	AIDE	NVIDIA
Playground subset (13)	44% (last)	47%	58% (first)
Full baseline (18)	59% (first)	46%	45% (last)

Mechanism: the 5 non-Playground competitions are old or niche — no mature public kernels, so NVIDIA's reproduction strategy loses its precondition (PR drops to 18.8 on afsis). Median PR, in contrast, orders the three identically on both sets. **“Which agent is stronger” is incomplete — the question is *on what distribution of competitions*.**

10 Failure modes

- **AIDE: diagnosis cannot veto the score.** On s3e19 an unrelated bug fix silently switched validation to shuffled KFold: metric jumped 18.4 $\rightarrow$ 4.4. Controlled experiment: same features, same model, split changed only — **4.5 $\times$ optimism bias** (4.56 vs. 20.41). AIDE spent 9 steps tuning a leaked metric. my-agent hit the same trap but tagged it “DIAGNOSTIC: not for submission.”
- **NVIDIA: votes $\neq$ quality.** An 89-vote workhorse kernel beat all three agents; a 7,859-vote tutorial kernel came last. Nothing checks the reproduced score against the source's claimed rank.
- **Shared gap:** none of the three recognizes “this task needs external data” (s3e19 requires GDP data; all three landed in the bimodal leaderboard's lower mode).

11 Conclusions

1. A credible 3-way comparison on 18 competitions: my-agent 59% / AIDE 46% / NVIDIA 45%, with 26% of duels undecidable.
2. A **paired significance test** for agent benchmarks whose error scale comes free from Kaggle's public/private split.
3. Local CV disagrees with the real leaderboard in 10/13 competitions — CV-only agent benchmarks are unreliable.
4. Benchmark claims must state the **evaluation-set composition** and report both absolute (percentile) and relative (paired win rate) metrics.

12 Ongoing work

Between-run variance of AIDE (repeatability re-runs on the 5 closest competitions); extension to NLP / time-series / imaging competitions; validation gates that let diagnoses veto scores; experience-library ablation.

Code, experiment logs, and per-competition ML specification reports: github.com/930731haohao-afk/kaggle-agent (private; access on request). Every number is traceable to structured experiment records.